Journal of Energy Storage 67 (2023) 107602

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Research papers

The overarching role of electric vehicles, power-to-hydrogen, and pumped hydro storage technologies in maximizing renewable energy integration and power generation in Sub-Saharan Africa

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A R T I C L E I N F O

Keywords:
Variable renewable energy sources (VRES)
Decarbonization
Renewable energy penetration
Pumped hydro storage
Electric vehicles
Power-to-hydrogen
EnergyPlan

A B S T R A C T

Globally, efforts are underway to keep global warming well below 1.5–2 °C as outlined in the Paris Agreement. Ghana plans to achieve 10 % renewable energy (RE) participation in the national energy mix by 2030. In the Renewable Energy Masterplan 2030, the target is a total installed RE utility scale of 1094.63 MW. In the current work, we propose a better pathway to gradual decarbonization of the Ghanaian energy sector (power, transport, industry, residential, and others) at higher technical levels of variable RE (VRE) penetration, 0 critical excess electricity production (CEEP), and lower cost than the 2030 Business-as-Usual (BAU) scenario. In our proposed scenario, High Renewable Energy Penetration (HREP) 2030, we assess the overarching role of electric vehicle integration, power-to-gas (hydrogen), and pumped hydro storage technologies in maximizing VRE penetration in Ghana. The results from the current study show that it is technically feasible to include solar and wind power penetrations of 32.66 % and 38.11 % in the 2030 HREP scenario compared to the 2.64 % and 1.99 % of the 2030 BAU case, respectively. The HREP scenario will ultimately lead to 26.45, 30.86, and 23.34 % contribution of solar, wind, and thermal in the total electricity production in 2030 compared to the 2.36, 1.78, and 74 % of the BAU scenario, respectively. The CO_2 emissions and its cost in the HREP case are 22.5 and 22.81 % lower than in the BAU case, and the total annual cost of the former is 12 % lower than the latter. The total electricity production from wind and solar energy alone in the 2030 HREP will create an additional 723 jobs and 11.96 Mt avoided CO_2 emissions. The proposal made in the current study could usher in a long-term transition pathway that leads from the current fossil-based system to an affordable, efficient, sustainable, and secure energy future for Ghana. The findings presented in the current study should be of significance to decision-makers in formulating and adopting zero and low-carbon strategies for long-term decarbonization targets and planning in Ghana beyond 2030.

1. Introduction

Ghana has historically depended on large hydro for the country's power supply until after 2015 when the trend gradually shifted – and as of the end of 2020, about 70 % of the power generation was sourced from fossil fuel-based thermal power plants [1]. The continuous increase in the share of thermal generation in the country is a result of the government's effort to solve the perennial power supply challenges. Besides the power sector, Ghana's transport sector is another major source of excessive fossil fuel consumption in the country. As of the end of 2019, the total consumption of petroleum products by the transport sector reached nearly 3000 ktoe, representing about 80 % of the petroleum

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https://doi.org/10.1016/j.est.2023.107602
Received 8 December 2022; Received in revised form 17 April 2023; Accepted 1 May 2023
Available online 11 May 2023

products available for final energy consumption by all sectors in the country [1]. The carbon dioxide (CO_2) emissions by Ghana's transport sector in 2019 reached 9 Mt, an 80 % rise from 2010 levels [2]. The industrial and residential sectors are all culprits - with close to zero consumption of low- and zero-carbon fuels. With Ghana being a signatory of the Paris Agreement, the trend and rate in the type of fuel consumption across the country's energy sector needs urgent attention.

The country has recently updated its nationally determined contribution to the United Nations Framework Convention on Climate Change (UNFCCC), and the update covers over 19 policy areas. The ultimate goal of these policies is to generate greenhouse (GHG) emission reductions of 64 Mt CO_2eq, generate over one million jobs, and avoid 2900 deaths due to improved air quality by 2030 [3]. In addition, the country's Renewable Energy Master Plan (REMP) targets a 10 % renewable energy (RE) integration in the final energy mix. To reach this target, the government has made several proposals mostly related to increasing the installed electricity capacities of renewable energy sources (RES) (excluding large hydro) to 1363.63 MW, of which 1094.63 MW are utility-scale systems. The plan targets the creation of 220,000 jobs and avoided CO_2 emissions savings of 11 Mt by 2030 [4]. Considering the enormous solar and wind power potential in the country, these planned installed capacities of both resources (utility-scale 447.5 MW and 325 MW, respectively) are low, especially when there is a better approach to penetrating more of these RES at a cheaper cost with no technical challenges. By creating/introducing additional demands alongside storage systems, more variable renewable energy sources (VRES), could be penetrated, which offers a better pathway to gradual decarbonization of the Ghanaian energy system than the 2030 business as usual (BAU) path. Additional demands and storage in this context refer to technologies such as electric vehicle (EV) integration with and without vehicle-to-grid (V2G) capabilities, power-to-gas, and pumped hydro storage (PHS). One key setback with the higher installed capacities of VRES has to do with the generation of 'too' much excess electricity, referred to herein as critical excess electricity production (CEEP), that can neither be consumed locally nor exported to neighboring countries due to transmission line restrictions. The occupancy of the electricity network with this excess electricity can ultimately cause the electricity supply system not to work [5], therefore requiring energy curtailment. By making available additional energy demands (substitutes for fossil fuels) and storage options, higher levels of VRES can be penetrated, which reduces energy curtailment and the risk of system shutdown. In the current study, these options are introduced as changes to the 2030 BAU case of Ghana to assess their technical, economic, environmental, and social benefits.

Widespread EV adoption is a feasible solution to reduce fossil fuel consumption and CO_2 emissions from the transport sector [6,7]. According to Ref. [8], a 25 % EV integration in Ghana could lead to a reduction of gasoline consumption by 23 %, while Ampah et al. [9] have also recently shown that the replacement of 100 gasoline vehicles with EVs in Ghana has the potential to save 460 tons/year of CO_2 emissions. The gradual penetration of EVs to replace conventional transport fuels could create more power demand which gives room for penetrating more VRES [10]. However, if unregulated, the increase in energy demands from EVs could potentially exacerbate peaks and ramps in net load, creating more generation investment [11]. Besides, the simultaneous high penetrations of VRES and EVs on the grid network may cause a power imbalance between demand and supply, causing frequent voltage fluctuations, overload of powerlines and transformers, energy losses, and power quality degradation, which could cripple the electricity network's technical and economic performances [12,13]. One potential solution to alleviate the adverse combined effects of high penetrations of EVs is to consider V2G/G2V technologies. According to Robledo et al. [14], EVs are 95 % of the time parked, thus creating an opportunity for selling their power to the grid or during excess production of electricity, they can act as storage options to provide various services to grid networks such as high VRE penetration [15], reactive power support, active power regulation, tracking of VRES, load balancing, and current harmonic filtering. These technologies enable ancillary services, such as voltage and frequency control and spinning reserve [16]. Dumlao et al. [17] show the potential role of EVs in reducing curtailment in electricity grids with high solar photovoltaic (PV) penetration in Japan. It was revealed that without EV integration as mobile storage devices, the curtailment by 2025 could reach 20–25 %; however, this curtailment reduces to 10 % with the introduction of EVs.

The high penetration of VRES in an electricity network demands the availability of storage technologies for flexibly storing energy. The PHS is thus perceived as one of the optimal solutions for integrating high shares of intermittent energy sources, such as wind and solar, into the power grid [18,19]. As of 2020, global energy storage had reached 191.1 GW of cumulative installed capacity, of which the PHS accounted for at least 90 % [20]. The rationale behind the PHS is to use excess electricity to send water from a lower reservoir to an upper reservoir via a pump, and during peak demand, the opposite occurs through a turbine to generate electricity [21]. PHS provides a quick response to variations in electricity demand and supply in case of high shares of VRES, ensuring the stability of the network [18,22]. The PHS has the ability to provide the following services: bulk energy, high RE integration, ancillary, and transmission and distribution services. Either alone or combined, these services are valuable fundamentals to providing the needed flexibility for an electricity network [23]. Zhang et al. [24] showed that the curtailment of wind power declines from 20.4 % in a 40 GW wind power system with no PHS to 15.7 % when a 3.6 GW PHS is introduced – providing an additional 6.26 TWh of wind generated-electricity to be exploited which would have been otherwise curtailed. More of these overarching roles of PHS in penetrating high shares of VRES at reduced curtailments can be found in the following works Tuohy and O'Malley [25] and US DOE [26].

As already posited, a wide range of storage technologies, such as batteries, PHS, as well as compressed-air energy storage, can be employed to cope with the intermittent nature of VRES as they have the ability to partially provide the required flexibility. However, electricity storage presents high investment costs [27], and a single solution is not sufficient for providing the necessary load shifting – different solutions integrated together present a better opportunity for realizing the most out of VRES without creating imbalances on the electricity grid [28–30]. Alternatively, coupling the various sectors together with energy conversion can effectively provide system flexibility [30]. As a matter of fact, the power sector is usually less flexible than the other non-electrical energy sectors since the latter does not require an instantaneous balancing of demand and supply, as is the case in the former [28]. In addition, though PHS has long time energy storage capabilities, its global output of 170 GW is only a fraction of the total power generating capacity across the world – the Organization for Economic Cooperation and Development (OECD) countries alone have 3000 GW of total power generating capacity. Theoretically, PHS could be expanded, but in reality, the land requirement for this expansion complicates the prospect [31]. Therefore, there is a need to identify and integrate other options with long-term energy storage that could provide similar or superior advantages to the electricity network compared to the electrical storage options. Hydrogen production through power-to-gas (P2G) technologies offer another cost-effective means of integrating high shares of VRES [32], where the surplus electricity is used to split water into hydrogen and oxygen through electrolysis. The hydrogen can be consumed in other sectors as substitutes for fossil fuels [33], and the remaining could be stored in tanks or geological reservoirs or blended and integrated into the gas grid [34]. P2G thus provides more flexibility, larger scale, and longer-term storage, enabling high penetration of VRES and lower costs than other energy storage options [28,30,34,35]. More detailed discussions on the P2G concept and its advantages, especially on maximizing RE penetration, are found in the works of Zhang et al. [36], Mazza et al. [37], Li et al. [38], and Ozturk and Dincer [39]. Another critical challenge with direct electrification of the entire energy system by grid power or batteries is that sectors such as road trucking, heavy industry, international maritime and aviation require highly dense fuels and thus are hard-to-abate sectors. Hydrogen, ammonia, methanol, etc., are some of the fuels required in these sectors and through P2G concepts and sector interconnections, they can be produced using surplus power and delivered to the hard-to-abate sectors for consumption or stored for later use.

Based on the evidence suggested supra, it is beneficial to consider EVs (with smart charging), PHS, and P2G as part of Ghana's future energy system to create the enabling environment for increasing the share of RE installed capacity in the total installed power generating capacity. The above sets the foundation for the main motivation of the current study, as these aspects are hardly discussed in the existing studies for future energy transitions involving Ghana, as seen in Table 1. It is also important to stress that despite the lack of EVs (with smart charging), PHS, and P2G in the current energy system and that of 2030 BAU, the conditions available still provides enough room for more VRES, especially wind and solar, to be penetrated than it has been in the past or as planned by the government. Hence, one of the key objectives of the present study is to assess the 'optimum technical' penetration for solar and wind energy that could have been installed in 2019 or could be installed in 2030 mainly on the basis of CEEP and primary energy supply (in the absence of the planned introduction of the storage options). Several studies have provided key contributions to the future energy transition scenario of Ghana for achieving higher penetrations of RE. Table 1 summarizes the key contents of these studies.

Despite the significant contributions made by these studies in the space of Ghana's RE sector, there remain critical study gaps and questions about the sector left unanswered. These above-referenced studies are characterized by at least one of the following limitations:

(1) Lack of energy sector coupling/interaction between sectors. These studies are based on one sector, usually power or transport. The interaction of the power sector with other sectors is a more efficient and holistic approach to penetrating more VRES without causing technical challenges to the entire system [53].
(2) Lack of storage options such as the PHS, batteries, and P2G as part of the energy system analysis.
(3) Do not reveal the optimum technical penetration levels of VRES, such as wind and solar energy, that can be integrated into the Ghana energy system given the historic and future conditions in the absence of storage options.
(4) Do not reveal the holistic technical, economic, environmental, and social benefits of their results, especially against a BAU case.

Against this backdrop, the findings from the current study fill the above-mentioned gaps in future energy transition literature on Ghana in a holistic manner. From the given limitations, the main research questions addressed in the current study include the following:

- RQ1: What is the optimum technical penetration of solar power that could have been included in the national electricity mix in 2019?
- RQ2: What is the optimum technical penetration of solar and wind power that can be included in the national electricity mix in 2030 compared to the BAU planned capacities?
- RQ3a: What role can EV integration (dump charge), P2G (hydrogen), and biofuel consumption have in maximizing the technical penetration of VRES and their consumption?
- RQ3b: What role can EV integration (smart charge) and PHS have in maximizing the technical penetration of VRES without non-zero CEEP?
- RQ4: Will the proposed 2030 scenario in the current study be financially feasible and competitive against the planned 2030 BAU?
- RQ5: What are the consequential job benefits and avoided CO_2 emissions of the proposed 2030 scenario in the current study against the planned 2030 BAU?
- RQ6: How much of Ghana's land will be required to meet the high penetration levels of wind and solar energy proposed in the current study?

From the changes and introductions proposed in the current work, we anticipate a better Ghanaian future energy system (than the BAU case), which is characterized by (1) higher shares of wind and solar energy than fossil fuel-based power generation, (2) lower costs, (3) lower GHG emissions and higher avoided CO_2 emissions, and (4) more job creation. Being aware that the government of Ghana cannot implement all the suggestions of our proposed plan, the main purpose of the current study is not to call for the replacement of the 2030 REMP but rather to reveal the importance of EV integration (with V2G capabilities), PHS, and P2G from a technical, economic, environmental, and social point of view. This should set the basis for their implementation in

Table 1
Existing studies on Ghana's future energy system.

Ref.	Sector analyzed	Objective	Aspects of energy system analyzed	Storage considered
[40]	Power	Role of biomass in Sub-Saharan future power sector: Ghana as a case study	Technical and environmental	Yes
[41]	Power	Energy transition in Western and Central Africa: Ghana as part of focus countries	Technical and environmental	No
[42]	Power	2010–2040 pathways for power generation in Ghana	Technical, economic, environmental	No
[43]	Power	2018–2048 pathways for power generation in Ghana	Technical, economic, environmental	No
[44]	Power	2010–2040 pathways for power generation in Ghana	Technical, economic, environmental	No
[45]	Power	2010–2040 pathways for power generation in Ghana	Technical, economic, environmental	No
[46]	Power, transport, residential	Prospects of bioenergy in Ghana's future energy system	Technical and environmental	No
[47]	Power	2013–2030 pathways for power generation in Ghana: role of solar energy policies	Technical and environmental	No
[48]	Transport	2019–2035 pathways for biofuel integration in Ghana's transport sector	Technical and environmental	No
[49]	Power	2010–2050 pathways for power generation in Ghana	Technical, economic, environmental	No
[50]	All excluding transport	2015–2030 transition towards sustainable energy systems in Ghana	Technical and economic	No
[51][a]	Power	2010–2040 long term modelling of electricity production and consumption in Africa: Ghana as part of the total assessment	Technical and environmental	No
[52][a]	All	2010–2040 long term modelling of energy supply and demand in Africa: Ghana as part of the total assessment	Technical and environmental	No

[a] Difficult to tell the individual results for Ghana.

longer future planning since there is currently no official REMP beyond 2030 for Ghana. The objectives of this study have been achieved using EnergyPlan, a popular and powerful tool for modelling future energy transitions.

The present study adds to the existing and ongoing studies on global energy transition pathways. Contributions made in this study are applicable to many developing countries, especially countries from Africa that have relatively low RE targets in the next decade or two despite the abundance of these RES. For such a fast-growing region, the introduction of P2G, PHS, and EVs (with V2G capabilities) technologies across major energy-consuming sectors in the continent (and other developing regions outside the continent) creates a more enabling environment for reaching 100 % RE target globally.

2. Methodology and model validation

2.1. EnergyPlan

As an energy system analysis tool, EnergyPlan is a computer-based model that has continuously been developed since 1999 by Professor Henrik Lund and colleagues at Aalborg University, Denmark. The tool is a deterministic computer model that considers a bottom-up approach in dealing with energy transition pathways. It identifies feasible energy system optimizations through hourly computations at a fixed period [5]. Due to the availability of energy storage systems, EnergyPlan has the ability to cover a wide range of RES penetration [53]. Specifically, different types of storage systems, including short-term (battery and pumped hydro), medium-term (compressed air energy storage, high/medium temperature thermal energy storage) and long-term (gas storage such as power-to-gas technology), can be modelled in EnergyPlan.

As inputs, the tool considers energy demands (both renewable and fossil) across several sectors, i.e., power, cooling, heating, industry, and transport. The capacities, costs, and efficiencies/capacity factors of the technologies are also specified as inputs. Outputs of the model include energy balances, yearly energy productions, fuel consumptions, total primary energy supply (PES), total costs, emissions, etc. The tool also works based on two types of optimization strategies; technical and market economic optimizations. The former is selected when the user has the primary goal of minimizing electricity imports or exports and identifying the solution with the least fuel demands while at it. On the other hand, the latter seeks to identify the most cost-friendly solution on the basis of the business economic cost of each production unit [5]. Due to the combination of different inputs, there is the likelihood of errors in the model that may create problems in its real-life application. EnergyPlan thus provides warnings in such instances, and the tool has documentation explaining ways to deal with each warning. Warnings include critical excess electricity production, grid stability problem, power plant/import problem, syn/biogas shortage, V2G connection too small, low electrolyzer capacity, and so on. More details about the tool and its functionalities are explained in Ref. [5]. Fig. 1 is a layout of the EnergyPlan simulation tool.

2.2. Research model

In the use of EnergyPlan, some key assumptions and model optimizations have to be made. As a key input, the hourly distribution profile of VRES is required in EnergyPlan. In the current study, the distribution profile for the VRES for 2019 used for the calibration and validation of the model was the same profile adopted for 2030 scenarios. Based on the distribution profile of the VRES and the correction factors, the capacity factors for the power plants were estimated. Next, for the thermal power plant in the Ghana 2019 EnergyPlan model to produce the same electricity as that of 2019, a fixed import/export was included. Also, a maximum import/export transmission line capacity was estimated based on the actual electricity export in 2019. Furthermore, to ensure an accurate working of the model, EnergyPlan produces key warnings to

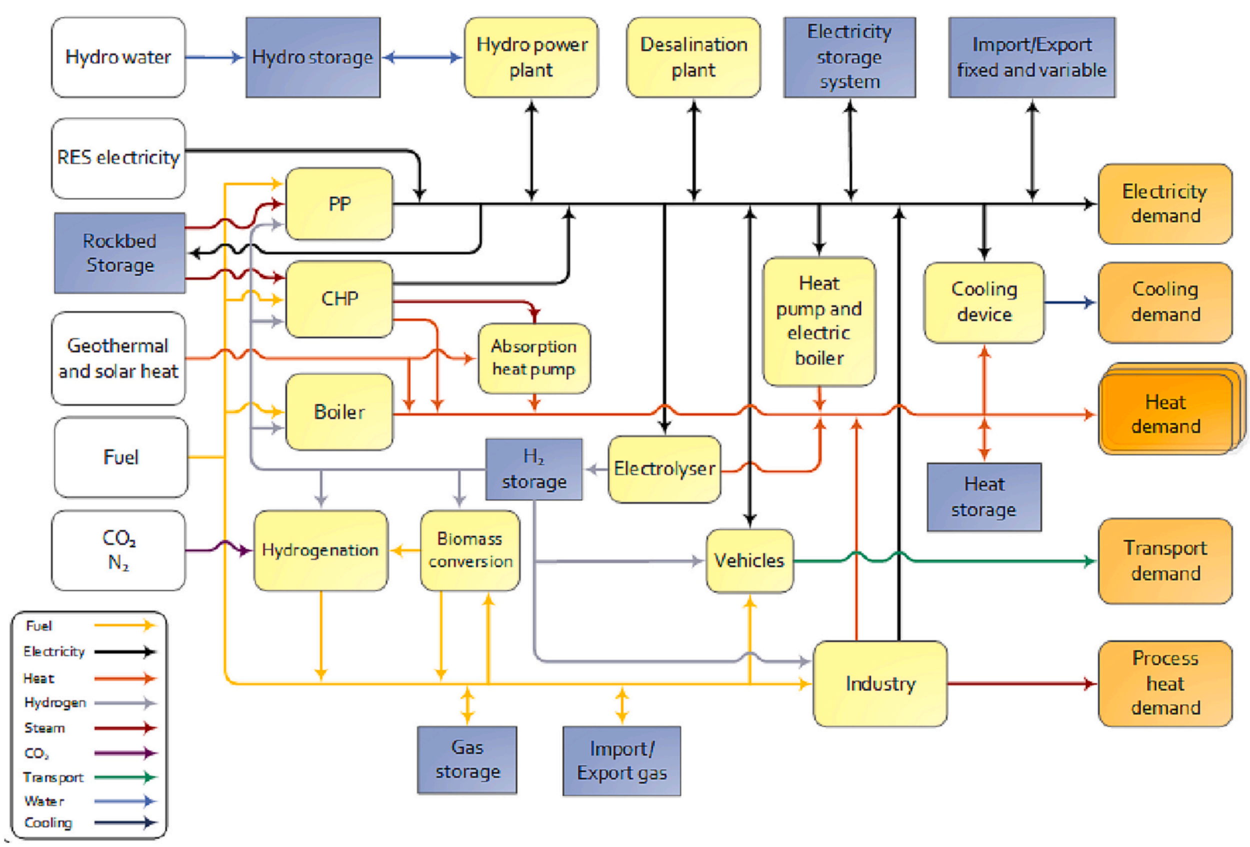


Fig. 1. General overview of the EnergyPlan tool [5].

advise users on potential problems in the model. Of particular interest are the CEEP and power production import (PPI) problems. The former is discussed in more detail in subsequent sections, but the investigators chose not to address the CEEP warning as it is critical in analysing the optimum technical penetration of solar energy and wind power in the Ghana electricity supply mix. However, with the PPI problem, the warning is a result of supply being lower than demand. In this instance, EnergyPlan imports the extra required electricity – however, if the import capacity during one or more hours exceeds the transmission line capacity, the PPI warning appears. Two options are available for addressing this warning - either by increasing the capacities of the power plants or the transmission line [5]. Since there were no actual distribution profiles for EV charging and fixed import/export for Ghana at the time of this investigation, the default profiles in EnergyPlan were adopted. In the current study, unless stated otherwise, we have used EnergyPlan's market economic optimization.

2.2.1. Scenario settings

The current study considers three main scenarios; the reference 2019 scenario (2019 REF), the business-as-usual 2030 scenario (2030 BAU), and the investigators' own suggested scenario for 2030, known as the 2030 High Renewable Energy Penetration scenario (2030 HREP). The 2030 HREP has been purposely designed to include innovative strategies such as electric vehicles with and without smart charging capabilities, pumped hydro storage, biofuel, and hydrogen production for high penetration of renewable energy in the 2030 Ghana energy mix. The subsequent subsections throw more light on the various scenarios.

2.2.2. 2019 REF scenario

In order to develop a comprehensive description of the existing energy structure of Ghana's energy sector and provide a reliable reference for future scenarios, the 2019 REF scenario was modelled. Considering 2019 as the base year, the model is developed expansively by taking power, industry, transport, and other energy-consuming sectors into account. 2019 has been selected because it is the initial year for Ghana's Renewable Energy Masterplan, which runs through 2030. The 2019 base data for the various sectors was adopted from the National Energy Statistics (2000–2019) [1] and Energy and Supply Outlook for Ghana [54], which contains actual data related to fuel consumption for power production, installed capacities, electricity production, electricity imports, and exports. The Strategic National Energy Plan (SNEP 2030) [8] was also a reference for the distribution of energy consumption in the transport sector by fuel type based on the 2020 projections. In this scenario, the actual penetration of VRE (solar power) as of 2019 is analyzed, after which the optimum technical penetration of solar power is investigated to ascertain how much solar power could have been added to the 2019 electricity mix without creating CEEP. The capacity utilization factors of the plants were taken from IRENA [55] assessment of power plants in Ghana as of 2020, i.e., hydro 52 %, solar 14 %, and bioenergy 27 %.

2.2.3. 2030 BAU

To create a BAU scenario, three official government reports were combined; Ref. [8] for the demand and Refs. [4,54] for the supply. Ghana's renewable energy target of 10 % RE share (excluding large hydro) in national electricity generation, which was previously set for 2020, had to be extended to 2030 since, by the end of 2019, the total RE electricity production was 51.3 GWh representing 0.3 % of the total electricity generated. In the 2030 Renewable Energy Masterplan of Ghana, one of the specific targets is to increase the share of renewable energy in the country's energy generation mix from 2015's 42.5 MW to 1363.63 MW, of which 1094.63 MW are utility-scale systems. For the current study, the installed capacities for the REs were based on the 2030 planned utility-scale capacities, whiles that of the thermal power plant and large hydro were kept at the 2019 levels. The capacities simulated are as follows; Solar PV (447.5 MW), wind (325 MW), medium/small hydro (150.03 MW), wave (50 MW), biomass co-generation (72 MW), MSW + biogas (50.1 MW), large hydro (1580 MW), natural-gas fired plants (3549 MW) (in recent years, the consumption of oil for power generation in Ghana continues to decrease – as such by 2030, we assume a 100 % natural gas-fired thermal power-plants). By 2030, the capacity factors for the RE plants for the 2030 BAU are as follows, medium/small hydro (54 %), solar (25 %), wind (26 %), wave (25 %), and bioenergy (27 %). For the demand side, the 2030 projections of energy demand across the power, transport, industry, and other sectors, as specified in Ref. [8], were adopted. In the 2030 models, the ratio of the fixed import to the total electricity demand and the ratio of the transmission line capacity to the total installed capacity was based on the ratios used in the validated 2019 model.

For the 2030 BAU, the model is first simulated based on the planned capacities, after which the optimum technical penetration of solar and wind energy is investigated based on the CEEP and PES. Following this, the combined optimum technical penetration of solar and wind energy is adopted for the building of the third scenario. This is to assess the influence of innovative strategies, such as PHS, EVs with G2V/V2G capabilities and hydrogen production via electrolysis, on limiting CEEP whiles penetrating more REs. The main purpose of the third scenario is to reveal how Ghana's energy sector can be made cleaner as a result of higher shares of RE compared to the BAU case and ultimately serve as a reference point for decision-makers during the planning of future energy targets.

2.2.4. 2030 HREP

This scenario builds on 2030 BAU by proposing several strategies to increase RE consumption in Ghana. The CEEP recorded in the combined optimum penetration of solar and wind power from the 2030 BAU is planned to be reduced to 0. In the 2030 HREP scenario, we expect a non-zero CEEP at both optimum technical penetration levels of wind and solar energy. To tackle the CEEP, several innovative strategies, mainly EV integration (dump charging and smart charging), hydrogen production (P2G), and PHS, are proposed for the Ghanaian energy sector. To maximize RE consumption, biofuels and solar thermal technologies are included. All parameters for 2030 BAU are kept constant for the HREP scenario except the following:

- The introduction of solar thermal technology with one day of heat storage in households should lead to the reduction of biomass consumption by approximately 30 %.
- 10 % of 2030 BAU fuel consumption in the industrial sector is replaced with hydrogen.
- Biogas for power production should increase from 2019's 0.0003 TWh to 0.3 TWh in 2030.
- The Ghanaian transport sector is mainly made up of passenger transport, freight transport, and bunkering, of which passenger transport fuel consumption makes up 71.95 %, which primarily constitutes light vehicles. In the current scenario, we propose 25 % of urban passenger vehicles (private cars, taxis, mini/midi buses) that run on gasoline are replaced with electric vehicles. The EVs are assumed to be equipped with 30 kWh lithium-ion batteries. Based on the information given, the electricity demand for the introduction of these EVs (approximately 1 million EVs) was 1.8 TWh/year, and the minimum required capacity of grid-to-battery connection found suitable for these EVs was 740 MW according to several iterations in EnergyPlan. The maximum share of cars during peak demand, the share of parked cars grid-connected, and efficiency (grid-to-battery) are considered as 20 %, 70 %, and 90 %, respectively. The rationale behind the inclusion of EVs is to allow for the vehicles to provide storage, matching the time of generation to the time of load [56]. In our scenario, all EVs are assumed to provide V2G services (in the intelligent charging case). This assumption is, however, hypothetical as it requires huge financial investment and time to realize – but the assumption made here helps to reveal the maximum possible effects

from V2G systems. Next, 20 % of diesel used in freight transport is replaced with biodiesel, and 3 % of intercity road transportation with diesel is replaced with hydrogen. In total, for the hydrogen requirement in the industry and transport sector, an electrolyzer capacity of 343 MW with an efficiency of 80 % was considered appropriate.

- In the work of Bamisile et al. [57], it is shown that the integration of V2G with PHS is more sustainable than V2G only as it gives more flexibility to the owners of the vehicles, and also PHS is a better storage technology than stationary battery storage in terms of sustainability and feasibility. As such, in the 2030 HREP scenario, we also consider a PHS with pumping/discharging efficiencies of 80 % and 33 GWh of storage capacity.

The authors will like to stress that most of these plans will likely not be adopted before 2030 (from a government's prioritization point of view not cost), considering we are only seven years away at the time of the current investigation. However, these strategies have been proposed in the current study with the sole aim of showing their advantages in increasing the optimum technical penetration level of VRES while at the same time avoiding a non-zero CEEP. The results from this scenario are intended to serve as a motivation for Ghanaian decision-makers in formulating plans for RE integration in the national energy mix in a more long-term future, such as 2050.

Fig. 2 shows the Ghanaian 2019 and 2030 energy models adopted in the current study. Also, Table 2 shows the various supplies and demands. The cost assumption data (Table 3) used in this study are based on the 2020 and 2030 cost databases provided by EnergyPlan developers [5]. Table 4 presents the prices of fuels in 2019 and 2030. The remaining cost types, such as variable O&M and cost for handling fuel (distribution and refinery) used in this study, can be found in the cost database of EnergyPlan 2020 and 2030. For both 2019 and 2030, an interest rate of 3 % has been used in the financial calculations of the current study. Though Ghana currently does not have carbon pricing, a CO_2 price of 28.6 and 34.6 EUR/t has been included in the current study for 2019 and 2030, respectively, to assess the influence of such pricing mechanisms in the Ghanaian energy sector going forward.

2.3. Model validation

Similar to any energy-modelled study, it is imperative to test the accuracy of a prepared model before applying it for further analysis in order to ensure robustness in findings. As such, a deviation between future actual and future modelled data could be minimized as high as possible. EnergyPlan-based studies use various methods to validate models. Some researchers compare actual data to modelled data by examining CO_2 emissions. Others look at differences in monthly electricity demand, and some have even tested models based on fuel consumption and electricity production. In this study, we validate our 2019 model by comparing it to actual data from multiple sources, including electricity production by plants, fuel consumption, the share of renewable energy sources in electricity production, electricity exports, and CO_2 emissions. By including multiple forms of data validation, we aim to increase the accuracy and reliability of our model. For the modelled thermal power plants and electricity exports to produce similar values, a fixed import/export value of 2.83 TWh and transmission line capacity of 166 MW was specified for the 2019 model, which was subject to change in the 2030 models. These additions to the 2019 model led to a total electricity demand of 16.77 TWh, of which 13.94 TWh was the net electricity consumption in Ghana as of 2019.

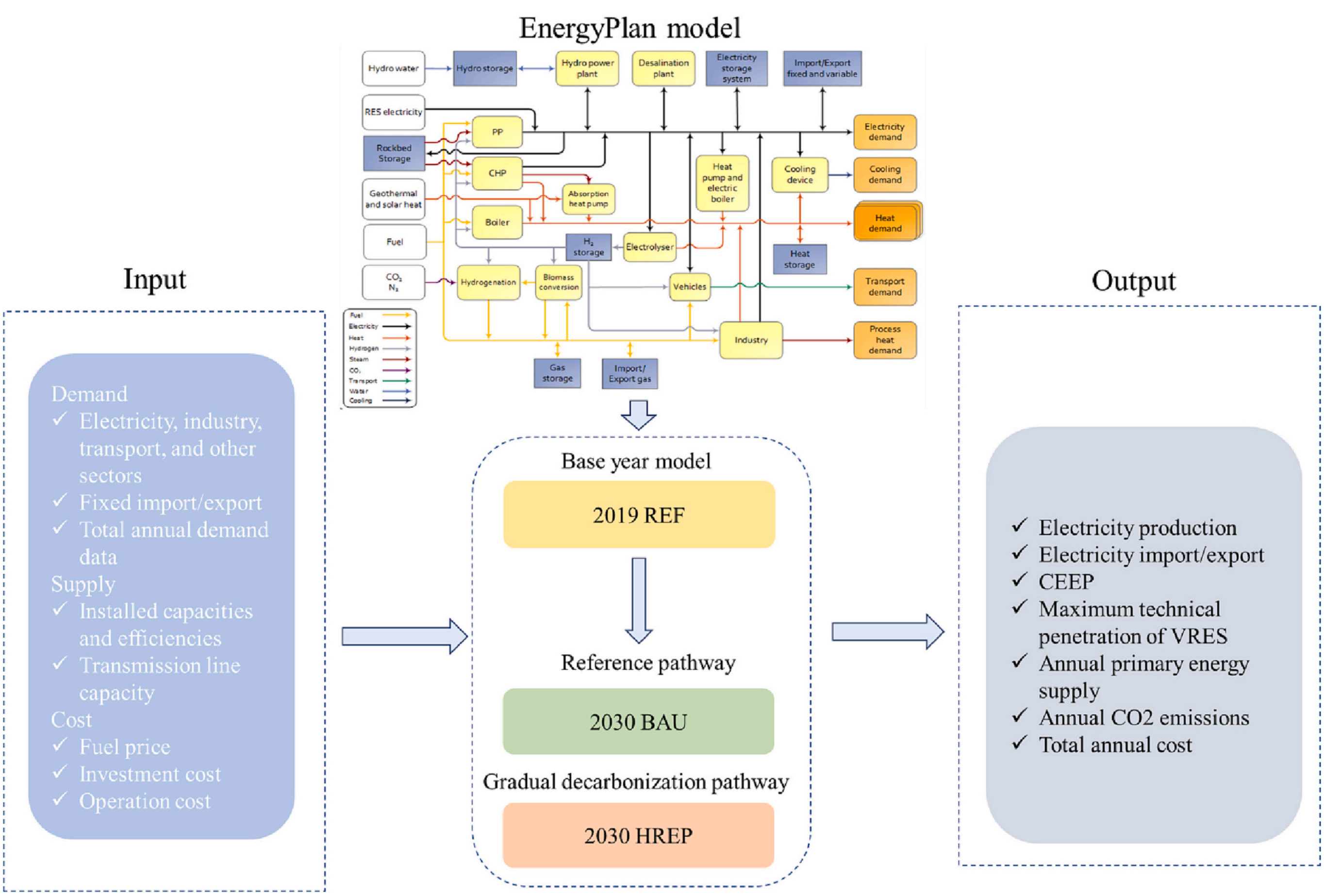


Fig. 2. Framework of Ghana 2019 and 2030 energy transition pathway in the current study.

Table 2
Energy demands and supply considered in the current study for 2019 and 2030 scenarios [1,4,8,54].

	2019 REF	2030 BAU
Demands (TWh)		
Power		
Electricity demand (excluding transport)	13.932	30.5
Electricity demand for transport	0.012	0.043
Industry		
Industrial fuel use (excluding electricity)		
Natural gas	0.85	1.5
Petroleum products	4.57	11.93
Biomass	3.33	6.66
Transport		
Transport fuel use (excluding electricity)		
Diesel	19.73[a]	58.31
Gasoline	12.16[a]	28.31
Jet kerosene	1.87[a]	4.55
LPG	0.56[a]	0.97
Other sectors' fuel use (excluding electricity)		
Petroleum products	4.38	9.86
Biomass	31.35	28.7
Supply[b] (MW)		
Power		
Thermal power plant	3549	3549
Biomass co-generation	0	72
Biogas + MSW	0.1	50.1
Large hydro	1580	1580
Wave power	0	50.1
Small/medium hydro	0	150.03
Solar	42.5	447.5
Wind	0	325

[a] Based on total demand from Ref. [1], the distributions according to fuel type were assumed based on the 2020 distributions in Ref. [8].
[b] Only utility scale.

Table 3
Cost assumptions adopted for financial analysis of 2019 and 2030 scenarios [5,58].

Plant/technology	2019 (MEUR/unit)	2030 (MEUR/unit)	Period (years)	O&M (% of investment)
Thermal power plant	0.99	0.98	27	3.12/3.16
Interconnection	1.2	1.2	40	1
Solar PV	1.02	0.82	35/40	1
Hydro	3.3	3.3	50	2
Boiler	5.8	5.8	21	2.6
Wind		1	25	2.59
Waste CHP	215.6	215.6	20	7.4
Wave		3.35	25	1
Biodiesel plant		2.53	20	3
Electrolyzer		0.35	15	3
Solar thermal		1533	30	1.22
EVs battery		100	12	5
EV charging station		[a]	13	0
Pumped hydro storage				
Pump		1.19	50	1.5
Turbine		1.19	50	1.5
Storage capacity		7.5	50	1.5

NB: values before and after '/' are for 2019 and 2030, respectively. Blank spaces imply the technology/plant was not considered for the year.
[a] For the work of Ref. [58], the total investment cost of a charging station for 3 million vehicles was 10% of the total investment of the Lithium-ion BEVs. Thus, we consider the same approach to estimate the total investment cost of the charging stations for the BEVs considered in this study.

Table 4
Cost assumptions for fuel prices used in 2019 and 2030 scenarios [5].

Fuel	2019 (EUR/GJ)	2030 (EUR/GJ)
Fuel oil	11.9	13.3
Diesel	15	17
Petrol/JP	16.1	17.6
Natural gas	9.1	10.2
LPG	17	18.7
Waste	0	0
Biomass	6.2	6.8
Biomass	6.2	6.8
Dry biomass	4.7	5.2

Table 5
Model validation results.

	Actual data	Modelled data	% Deviation
Production (TWh)			
Solar PV	0.05	0.05	0
Hydro	7.25	7.25	0
Thermal plant	10.88	10.88	0
Biogas power	0.0003	0.0003	0
Total electricity production	18.18	18.18	0
Consumption (TWh)			
Petroleum products	48.87[a]	48.80	−0.14
Natural gas	17.29[a]	17.08	−1.2
Biomass	34.67	34.68	0.02
Total consumption	99.20[b]	100.56	1.37
Others			
CO_2 emissions (Mt)	16.55[c]	16.50	−0.30
Share of RES (includes large hydro) in electricity production (%)	40.2	40.2	0
Electricity exports	1.43	1.43	0

[a] Sum of fuel consumed for power generatiom and use in other sectors.
[b] Calculated form Ref. [1]; final energy consumed – electricity + (crude oil, natural gas, and petroleum products consumption for electricity generation).
[c] Calculated based on the fuel consumptions and the CO_2 content of the fuels.

From Table 5, it can be seen that the percentage deviation between actual and modelled data ranges between −0.30 to 1.37 %, which we find acceptable and build upon to analyse Ghana's energy supply and demand outlook by 2030.

3. Results and discussions

3.1. 2019 REF

Fig. 3 shows the hourly distribution of electricity consumption in Ghana as of 2019. The annual average, minimum, and peak consumption according to the model were 1586, 1, and 2093 MW, respectively. A total of 18.18 TWh of electricity were produced domestically for consumption, of which 39.9 %, 59.8 %, and 0.3 % were produced by hydro, thermal, and renewables, respectively. In the current model for 2019, the electricity imports and exports were 0.03 and 1.43 TWh, respectively. Fig. 4 shows the hourly electricity production, imports, and exports. According to the 2019 model, the total CO_2 emissions based on the various fuel consumptions was 16.50 Mt.

3.1.1. *What is the optimum technical penetration of solar power that could have been included in the national electricity mix in 2019?*

The current question has been asked on the basis of the excessive under-utilization of solar energy for power production in Ghana. Considering the enormous potential of solar resource in Ghana, according to the 2019 model, the technical penetration of solar power in 2019 (given the total electricity demand) was only 0.29 %. This is a very

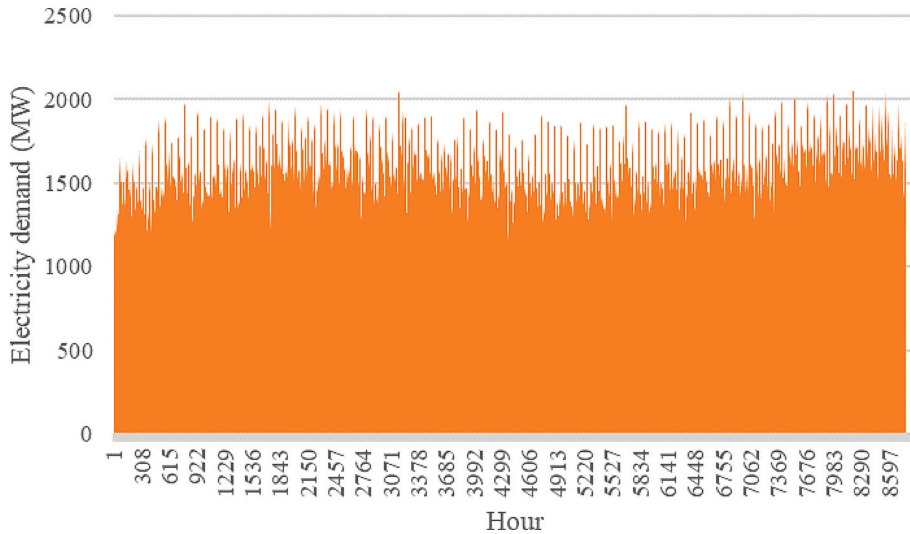


Fig. 3. Hourly electricity demand profile in Ghana based on the 2019 model (MW).

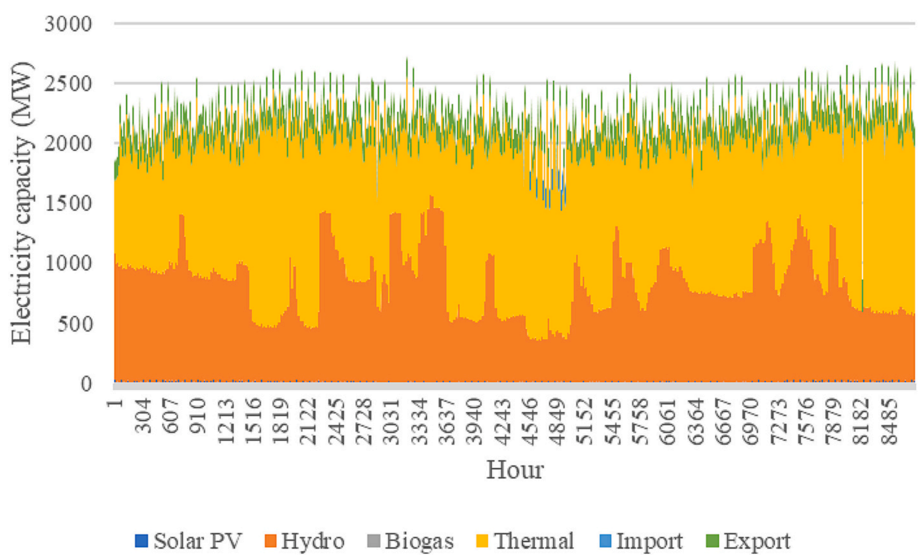


Fig. 4. Hourly electricity production, imports, and exports (MW).

abysmal penetration rate and justifies the country's plans to increase total solar installed capacity to 447.5 MW (utility-scale) by 2030. We answer the question asked from two angles; the total primary energy supply (TPES) and CEEP. Eq. (1) shows the calculation for VRE penetration (%) [59];

$$VRE\ Penetration\ (\%) = \frac{Total\ amount\ of\ VRE\ electricity\ produced}{Gross\ annual\ electricity\ demand} \tag{1}$$

$$Gross\ annual\ electricity\ demand = Total\ electricity\ generation + imports - exports \tag{2}$$

By increasing the penetration levels of VRES, the tendency for CEEP to increase above 0 becomes more likely. In a power network, the CEEP is the excess electricity that can neither be consumed locally nor exported as this excess during one or more hours exceeds the transmission line capacity [5]. The problem requires energy curtailment since an excess of electricity in the system has the potential to cause the electricity supply system not to work. In such an event, the electricity supply system lacks the capability to operate. This is a very typical phenomenon of high penetration levels of VRES and can only be resolved via storage technologies or expansion of transmission line capacity with neighboring countries to increase export levels [5]. Hence, it is critical to investigate the optimum technical RE penetration that can be included in the system at which CEEP remains at 0.

However, it is worth stating that the increase in RE penetration has the tendency to lead to a reduction in fuel consumption to some extent which is a positive feature for the reduction in fuel consumption cost and CO_2 emissions. Therefore, it is also fair to assess the optimum technical penetration of VRES for which fuel consumption (PES) reduces to its lowest level. Since CEEP and PES can both give an indication of the optimum technical penetration of VRES, Connolly et al. [60] developed the compromise coefficient (COMP). The COMP is the ratio of the reduction in PES to the increase in CEEP in each simulation (Eq. (3)) [60];

$$COMP = \frac{\Delta PES}{\Delta CEEP} \tag{3}$$

The COMP represents the benefits of adding VRES (i.e., reducing consumption of fuel) against the drawbacks of adding VRES (i.e., consequently leading to CEEP) [60]. Hence, the maximum feasible penetration of a particular VRES occurs when COMP is <1 [60]. To

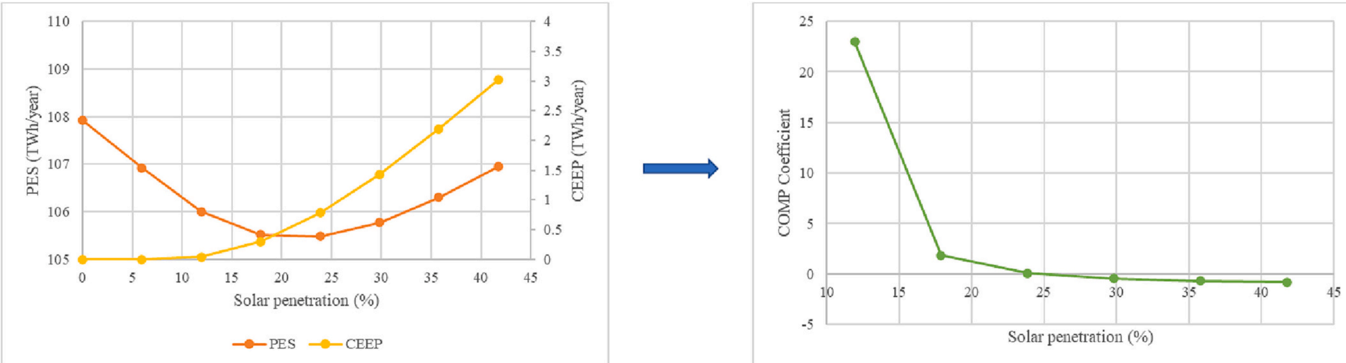


Fig. 5. Optimum technical penetration of solar power in 2019 based on CEEP and TPES.

identify the optimum technical penetration of solar energy in the 2019 model, all parameters are kept constant, and the solar electricity production is increased gradually at intervals of 1 TWh. For every 1 TWh change in electricity production by solar energy, the PES and CEEP are estimated, after which the various COMPs are calculated.

As seen in Fig. 5, from a technical penetration of 0 to 5 % for solar energy, the CEEP is 0, and the trend continues until after 8.7 %, after which CEEP begins to increase above 0. Hence, at 8.7 %, the maximum technical penetration of solar energy is reached. CEEP, however, does not indicate the optimum technical penetration but only indicates the minimum boundary, as, at this point, CEEP is still 0 [61]. Hence, the COMP is calculated further for the maximum penetration boundary. As seen in Fig. 5, from a solar penetration of 0 to ~24 %, fuel consumption decreases with an increase in solar penetration, after which the consumption starts to increase gradually. At this point, both CEEP and PES are increasing, and COMP starts returning negative values. Hence, with regards to PES, the maximum penetration of solar power in the 2019 Ghana electricity mix that could have been used is 23.85 %. Based on the two results from CEEP and PES, the 'optimum technical' penetration of solar power that could have been included in the 2019 electricity mix of Ghana lies between 8.7 and 23.85 % as opposed to the actual 0.29 % penetration in 2019.

3.2. 2019 REF costs

The 2019 energy system based on our model costs a total of 5899 MEUR, of which the largest share comes from variable costs, followed by annual investment costs and fixed operating costs. Though Ghana does not have carbon pricing, the total annual cost above includes CO_2 emissions cost of 472 MEUR, which is 8 % of the total cost. As the world continues to strive for carbon peak and neutrality, Ghana could adopt a carbon pricing scheme in the future as one of the strategies to promote clean fuel consumption. The results obtained herein go to show the economic importance of phasing-out carbon-based fuels and their technologies. Given the 8.7–23.85 % optimum penetration of solar power that could have been included in 2019, the total cost of an 8.7 % solar penetration energy system is 5886 MEUR with a CO_2 emissions cost of 454 MEUR, which is 0.22 % and 3.8 % lower than the 0.29 % actual solar penetration in 2019. Also, for the 23.85 % solar penetration, the total annual cost of the system is 5942 MEUR which is only 0.72 % higher (compared to the base case) due to the significant increase in solar PV cost, but the CO_2 emissions cost is 8.47 % lower. It is also worth noting that the fuel consumption/handling and natural gas exchange costs decrease with an increase in penetration rates. At 0.29 %, 8.7 %, and 23.85 % technical solar penetration, these costs are 4579, 4476, and 4346 MEUR, respectively. The full breakdown of the cost characteristics of the 2019 energy system according to different technical penetrations of solar energy is shown in Fig. 6.

3.3. 2030 BAU

In the 2030 BAU scenario, the annual average, maximum, and minimum electricity demand for the annual net electricity demand of 30.50 TWh (excluding the transport sector) were 3472, 4582, and 3 MW, respectively. The total electricity production will reach 41.07 TWh, and 0.88 TWh and 5.19 TWh of electricity imports and exports will be

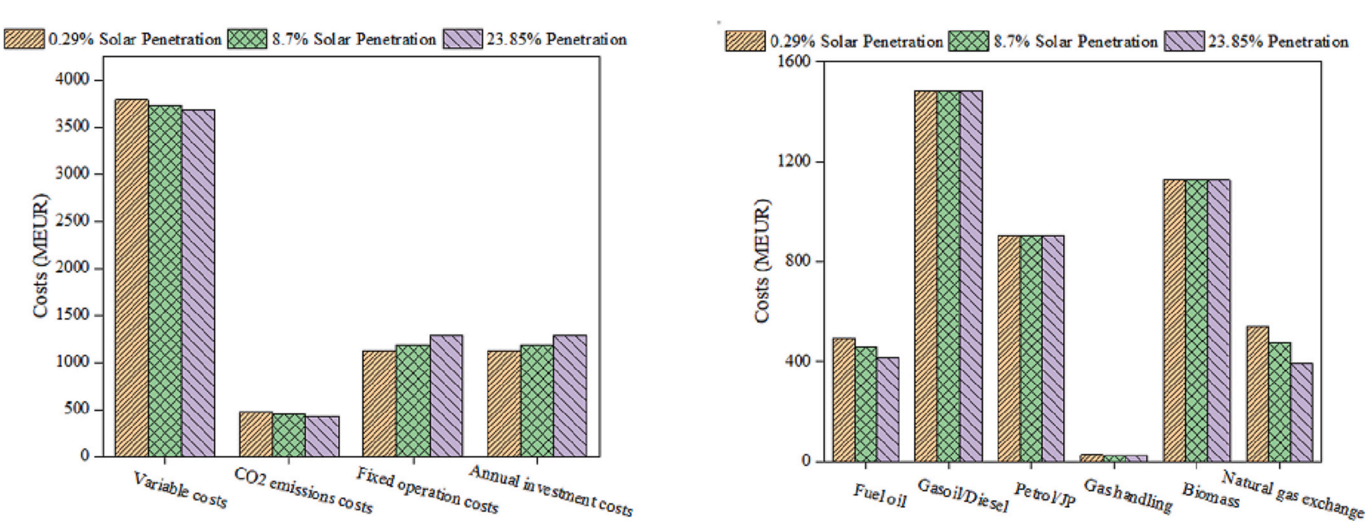


Fig. 6. Cost characteristics of the 2019 energy system according to different technical penetrations of solar energy.

recorded, respectively. According to the 2030 BAU model, thermal electricity generation will continue to dominate as it has been since 2016, albeit much higher share in 2030 compared to 2019. The 2030 BAU model shows that electricity generation from thermal generation will constitute about 74 % of the total, followed by large hydro with 18 % and renewables with 8 %. Though this 8 % is significant compared to the 0.3 % in 2019, we show later in this section that much more renewables, mainly solar and wind energy, can be penetrated in the 2030 energy mix without non-zero CEEP. Due to the increase in energy demand from 2019 to 2030, the total fuel consumption (excluding RES) increases to 214.67 TWh/year as opposed to 100.56 TWh/year in 2019, representing a 113.47 % increase. The increase in fuel consumption will lead to CO_2 emissions of 43.09 Mt.

3.3.1. What is the optimum technical penetration of solar and wind power that can be included in the national electricity mix in 2030 compared to the BAU planned capacities?

Based on the planned installed capacities (utility-scale), the technical penetration of solar and wind power will be 2.64 % and 1.99 %, respectively, by 2030. Despite the rise in penetration rates of these two VRES compared to 2019 levels, the simulation shows a CEEP of 0. Thus, there is more room for increasing the technical penetration of solar and wind power than what has been planned for 2030 in the BAU case. This is to ultimately maximize RE consumption, reduce carbon-based fuel consumption, and reduce CO_2 emissions as high as possible. To assess the optimum technical penetration for solar and wind energy in the given 2030 conditions, two scenarios have been set up. As the capacity of one renewable energy increases, the capacity of the others remains at the 2030 BAU level. In simple terms, as solar capacity increases from 0 MW to its possible limit without non-zero CEEP, the capacity of wind power remain at 325 MW. On the other hand, as the capacity of wind power increases from 0 MW to its possible limit, the capacity of solar energy remains at 447.5 MW. In both instances, the installed capacities of all other energy sources remain at the BAU level. This approach is consistent with that of Ref. [61].

From Fig. 7, it is seen that for CEEP, the maximum technical penetration of solar energy that can be included in the 2030 energy mix is 32.66 %, as at this level, the total CEEP generated is still 0. On the other hand, for PES, this level increases to 51.7 % as this is the maximum level that fuel consumption reduces, after which it begins to increase gradually. The COMP coefficient at this point is 0.11, after which negative values begin to appear. Hence, the range of optimum technical solar power penetration for 2030 could lie between 32.66 and 51.7 % compared to the BAU case of 2.64 %. The corresponding solar electricity production for these penetrations lies between 12 and 19 TWh/year. In the same Fig. 7, it is seen that the range of optimum technical wind energy penetration for 2030 lies between 38.11 and 73.49 % compared to the BAU case of 1.99 %. The wind electricity production for these penetrations lies between 14 and 27 TWh/year. The results show that wind energy will have a higher penetration rate than solar power for electricity production in Ghana by 2030. Both Ref. [61] and Ref. [62] report similar results, and the latter explains that the longer availability of wind energy in the day compared to solar power's restriction to only daylight hours is the main reason for the better penetration of wind energy than solar power.

Tables 6–8 summarize CEEP, PES, and COMP for 2019 solar power penetration, 2030 solar penetration, and 2030 wind energy penetration in the Ghanaian energy system.

Based on the optimum technical penetrations of solar and wind energy based on both CEEP and PES, the share of electricity generation by renewables, large hydro, and thermal power was determined and compared against the penetrations in the BAU case (2.64 % solar and 1.99 % wind energy). Of note, the share of electricity generation from the thermal power plant in 2019 was 59.8 %. The dominance of thermal power for electricity generation in Ghana started in 2016 and increased yearly whiles the share of electricity from hydro continued to recede. Based on the energy demands in 2030, our model shows that the trend is likely to continue, with thermal generation contributing at least 74 % of

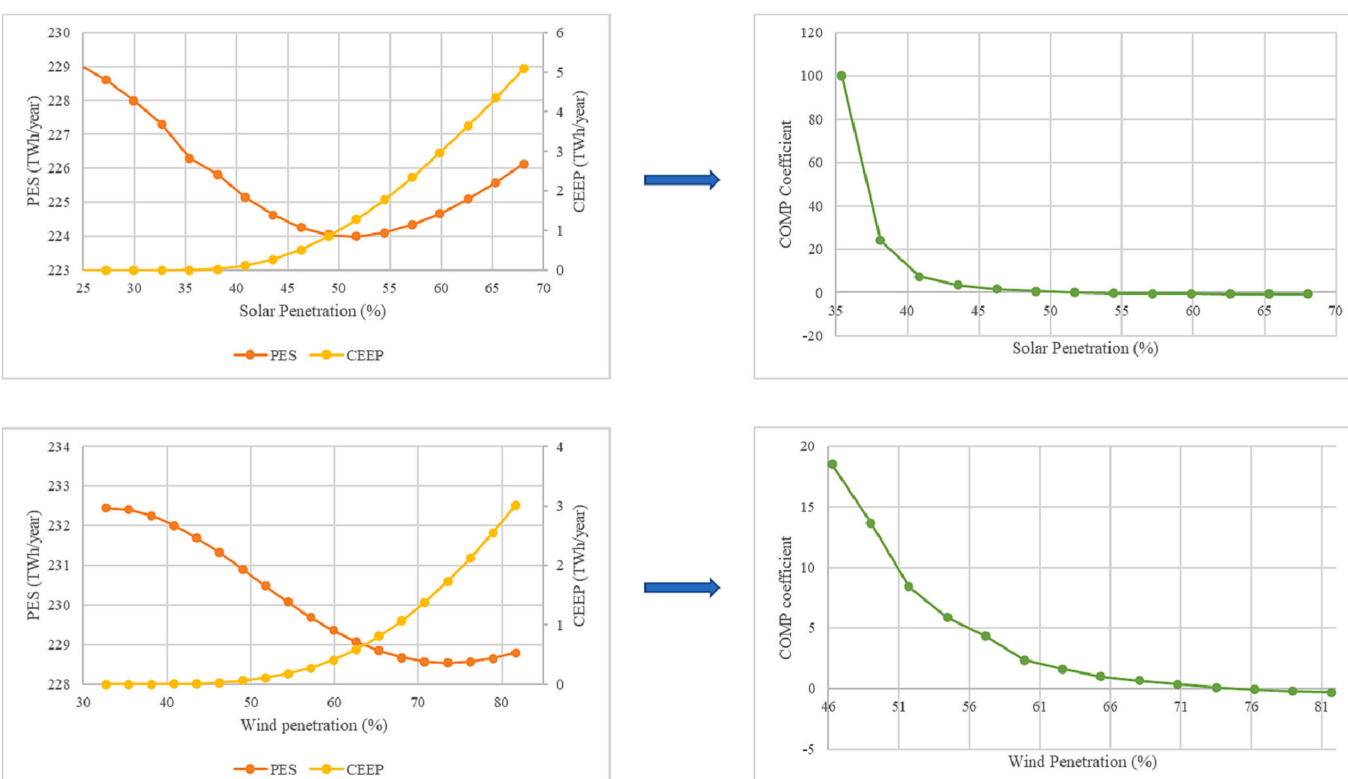


Fig. 7. Optimum technical penetration of solar and wind power in 2030 based on CEEP and TPES.

Table 6
CEEP, PES, and COMP for calculating optimum technical solar penetration for Ghana's energy system in 2019.

TWh	MW	% PEN	PES	CEEP	COMP
0	0	0	107.92	0	–
1	839	5.96	106.92	0	–
2	1681	11.92	106	0.04	23
3	2524	17.89	105.52	0.3	1.846154
4	3367	23.85	105.48	0.78	0.083333
5	4209	29.82	105.77	1.43	−0.44615
6	5052	35.77	106.3	2.19	−0.69737
7	5894	41.74	106.95	3.02	−0.78313

Note: % PEN refers to the penetration percent.

Table 7
CEEP, PES, and COMP for calculating optimum technical solar penetration for Ghana's energy system in 2030.

TWh	% PEN	PES	CEEP	COMP
0	0	223.47	0	–
8	21.77463	229.36	0	–
9	24.49646	229.08	0	–
10	27.21829	228.6	0	–
11	29.94012	227.99	0	–
12	32.66195	227.3	0	–
13	35.38378	226.3	0.01	100
14	38.10561	225.82	0.03	24
15	40.82744	225.16	0.12	7.333333
16	43.54927	224.63	0.27	3.533333
17	46.27109	224.26	0.52	1.48
18	48.99292	224.05	0.85	0.636364
19	51.71475	224	1.28	0.116279
20	54.43658	224.11	1.78	−0.22
21	57.15841	224.34	2.35	−0.40351
22	59.88024	224.67	2.97	−0.53226
23	62.60207	225.09	3.64	−0.62687
24	65.3239	225.58	4.35	−0.69014
25	68.04573	226.13	5.09	−0.74324

Table 8
CEEP, PES, and COMP for calculating optimum technical wind penetration for Ghana's energy system in 2030.

TWh	% PEN	PES	CEEP	COMP
12	32.66195	232.45	0	–
13	35.38378	232.41	0	–
14	38.10561	232.25	0	–
15	40.82744	232.01	0.01	24
16	43.54927	231.69	0.01	–
17	46.27109	231.32	0.03	18.5
18	48.99292	230.91	0.06	13.66667
19	51.71475	230.49	0.11	8.4
20	54.43658	230.08	0.18	5.857143
21	57.15841	229.69	0.27	4.333333
22	59.88024	229.36	0.41	2.357143
23	62.60207	229.07	0.59	1.611111
24	65.3239	228.85	0.81	1
25	68.04573	228.68	1.07	0.653846
26	70.76756	228.57	1.38	0.354839
27	73.48938	228.54	1.73	0.085714
28	76.21121	228.57	2.12	−0.07692
29	78.93304	228.66	2.55	−0.2093
30	81.65487	228.8	3.01	−0.30435

the total electricity, with 18 and 8 % from large hydro and renewables, respectively. By penetrating solar to its maximum penetration levels with 0 CEEP (32.66 % solar and 1.99 % wind), the share of electricity generation from thermal power reduces to 56 %, whiles renewables increase to 29 % with 15 % from large hydro (Fig. 8). In the same manner, by penetrating wind energy to its maximum penetration with 0 CEEP (2.64 % solar and 38.11 % wind), the electricity share from thermal, large hydro, and renewables are 54 %, 14 %, and 32 %, respectively. With regards to penetrations based on PES, maximum penetrations of solar and wind will lead to thermal and renewables (excluding large hydro) electricity generation of 43 and 42 % and 35 and 52 %, respectively. At 2030 BAU, the energy-related CO_2 emissions will be around 43.09 Mt, representing a 161.14 % rise from 2019 levels. At the solar and wind penetrations of 32.66–51.7 % and 38.11–73.49 %, the 2030 BAU emissions could reduce to 41.55–39.50 Mt and 42.11–38.81 Mt, respectively.

3.4. 2030 BAU costs

The financial characteristics of the 2030 BAU show that around 12,357 MEUR of the total annual cost will be needed to meet the energy demands of Ghanaians in 2030. This is approximately twice the cost of 2019 REF. From 2019 to 2030, the CO_2 emissions cost (if Ghana had/plans to implement carbon pricing) will increase from 472 MEUR to 1491 MEUR. Similarly, the cost of fuel consumption/handling/exchange of natural gas increases from 4038 MEUR to 11,302 MEUR. The significant rise in total annual cost from 2019 to 2030 could be attributed mostly to the relatively high fuel demand in the transport sector. Between 2019 and 2030, the energy demand of the transport sector increases by 168.33 % compared to 129.6 % and 118.8 % of the industry and power sectors, respectively. Fig. 9 represents the breakdown of the total cost of investment for different penetrations of solar and wind in the national energy mix by 2030. The 2.64 % solar and 1.99 % wind power penetration represents the BAU case. On the basis of variable costs (including CO_2 emission cost), the cheapest plan is the combined 2.64 % solar and 73.49 % wind power penetration. The BAU case, however, has the least costs for fixed operation and annual investment as the addition of more capacities of the VRES adds more costs to the other alternatives.

By looking at the costs of fuel consumption/gas handling/natural gas exchange, it is seen that the addition of VRES capacity only leads to a reduction in the costs of gas handling, biomass, and natural gas exchange. The 2030 scenario involves a 100 % natural gas-fired power plant; as such, fuel oil consumption remains unchanged as VRES capacities increase. Hence, with a focus on natural gas exchange, the associated costs reduce by 11.84 %, 28.07 %, 7.66 %, and 33.51 % compared to the BAU case for 32.66 % solar and 1.99 % wind power penetration, 51.7 % solar and 1.99 % wind power penetration, 2.64 % solar and 38.11 % wind power penetration, and 2.64 % solar and 73.49 % wind power penetration, respectively.

In the subsequent scenario, the combined optimum technical penetrations of both solar and wind power are investigated to reveal the importance of EV integration, smart charging of EVs, and hydrogen production via electrolysis on limiting CEEP at higher penetration of VRES. While at it, biofuel consumption in the transport sector and solar thermal technologies in the residential sector are included in the scenario to maximize RE consumption across all sectors.

3.5. 2030 HREP

From the previous scenario, the maximum penetration of solar energy (when penetration of wind is kept constant in the BAU case) was 32.66 %, where CEEP was still 0 TWh. Similarly, the maximum penetration of wind power (when penetration of solar energy is kept constant at BAU case) was 38.11 %, where CEEP was still 0 TWh. However, by combining both penetration levels simultaneously in the 2030 BAU case, CEEP increases from 0 to 0.61 TWh, which is a threat to the proper functioning of the entire power system as the entire electricity supply system could stop working with the occurrence of a non-zero CEEP. In order to maintain both penetrations of solar and wind power without worrying about its related CEEP problems, there is the need to create more demand and introduce storage systems into the existing framework. The scenario also considers the consumption of biofuels and the inclusion of solar thermal technologies in the transport and residential sectors, respectively. The specific plans for this scenario are highlighted

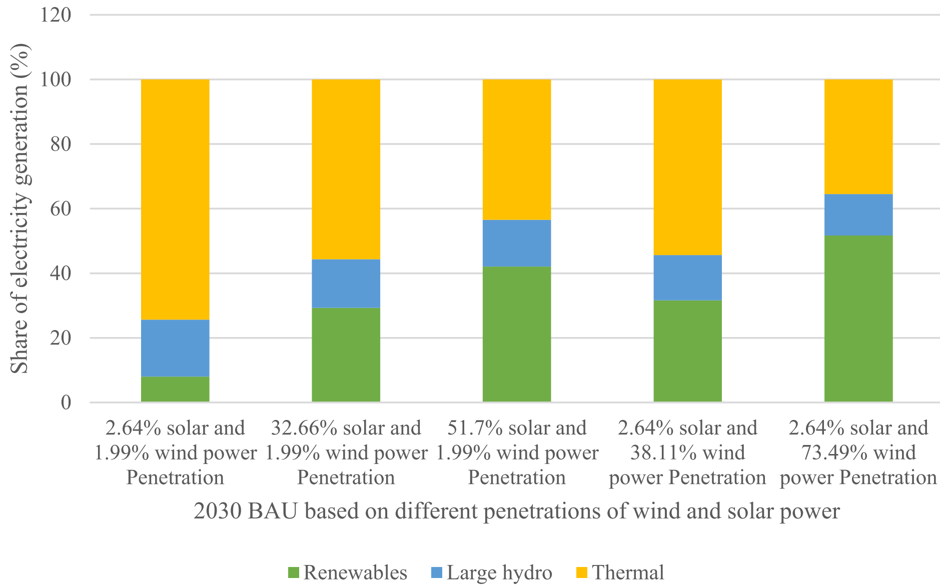
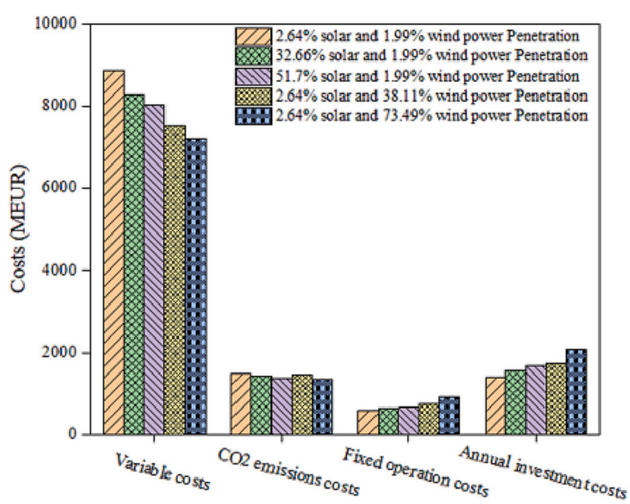


Fig. 8. Share of electricity generation in 2030 according to different levels of VRES technical penetration.



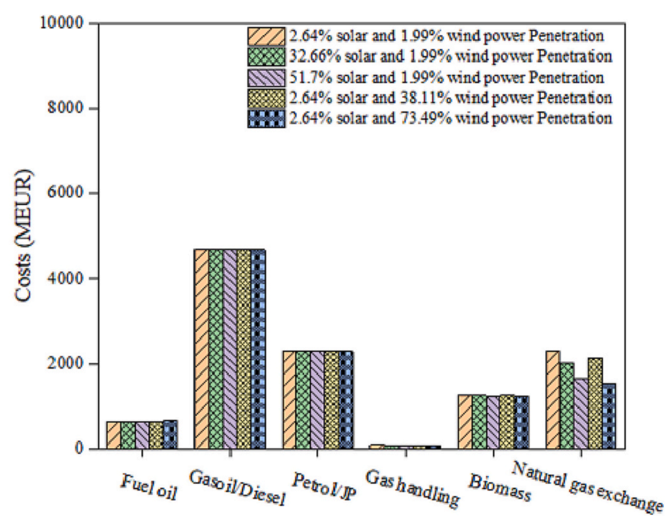


Fig. 9. Cost characteristics of 2030 energy system according to different technical penetrations of solar energy.

in Section 2.2.4, but the first three strategies (including dump EV charging) are investigated, after which V2G and PHS are included to ascertain their importance in the penetration of VRES at 0 CEEP.

As already pointed out, all of these proposals are likely not attainable by 2030 in Ghana, but the importance of the results obtained should serve as a guide for the Ghanaian decision-makers for long-term planning in the next two to three decades.

3.5.1. What role can EV integration, hydrogen, and biofuel consumption have in maximizing technical penetration of VRES and their consumption?

The first part of the results in this scenario excludes the smart charging of EVs (V2G) and PHS, as outlined in Section 3.5. It is worth noting that since there is a final plan to consider V2G and PHS for maximizing VRES penetration whiles avoiding CEEP (>0), the simulations in this section are based on the technical simulation (balancing both heat and electricity demands) in EnergyPlan.

Changes made to the 2030 BAU show that the inclusion of solar thermal technologies for the residential sector, increment in biogas for power, biofuels for transport, hydrogen for industry and transport, and electric vehicles (dump charge) with solar energy and wind power penetrations of 32.66 % and 38.11 %, respectively lead to a CO_2 emission of 31.91 Mt compared to the 43.09 Mt (market economic simulation) or 41.18 Mt (technical simulation) of the 2030 BAU case, representing a 22.5 % reduction (technical simulation case). RES electricity production increases to 34.78 TWh/year as against the 10.53 TWh/year of the 2030 BAU. The results obtained show the importance of the proposed strategies in maximizing RE consumption. The gradual introduction of EVs, biodiesel, and hydrogen in the transport sector ensures that CO_2 emissions from the sector reduce from the BAU case by 11.43 %; similarly, the inclusion of hydrogen in the industrial sector led to a reduction of CO_2 emissions from the BAU case by 8.91 %. Despite these benefits, the scenario has a CEEP of 0.47 TWh remaining in the system. It is worth noting that at these combined maximum penetrations of solar and wind power, the CEEP for the BAU case was originally 0.61 TWh – thus, the new introductions made in this scenario eliminated 23 % of the original CEEP. However, since the optimal target is 100 % elimination of CEEP, we test the effect of EV integration (smart charging) and PHS.

Table 9
Assessing the effect of EV (dump and smart charge) and PHS on CEEP, CO_2 emissions, and fuel consumption.

Strategies 1–3 of the HREP scenario with	CEEP remaining (TWh/year)	CO_2 emissions (Mt)	PES (TWh/year)
Dump charge of EVs	0.47	31.913	199.17
Smart charge of EVs (V2G)	0.26	31.851	198.85
Dump charge of EVs with PHS	0.05	31.842	198.80
Smart charge of EVs (V2G) with PHS	0	31.804	198.60

3.5.2. What role can EV integration (smart charge) and PHS have in maximizing the technical penetration of VRES without non-zero CEEP?

As seen in Table 9, the proposed strategies, in addition to EV integration (dump charge), show a CEEP of 0.47 TWh remaining in the system after including the maximum combined penetrations of solar and wind power from the 2030 BAU. The corresponding CO_2 emission and PES are 31.913 Mt and 199.17 TWh/year, respectively. If 100 % of the EVs are equipped with smart charging capabilities, CEEP reduces to 0.26 TWh, and the corresponding CO_2 emission and PES are 31.851 Mt and 198.85 TWh, respectively, a decrease of 44.68 %, 0.19 %, and 0.16 %, respectively compared to the dump-charging case. The results show that V2G is more beneficial for storage, matching the generation time to load time and consequently limiting CEEP at higher penetrations of VRES. However, based on the proposed plan (combined maximum penetrations of both solar and wind energy), the inclusion of V2G (at the selected battery capacity) was not enough to reduce CEEP to 0. By adding PHS, CEEP reduces to 0, and both CO_2 emissions and fuel consumption reduce accordingly. Thus, future energy transition pathways for Ghana could realize the most technical benefits if both smart charging EVs and PHS are simultaneously integrated. Based on the current findings, the remaining discussions relate to the scenario, including both EVs (smart charge) and PHS.

Compared to the BAU case (8 % share of electricity generation from RE excluding large hydro), the 2030 HREP scenario leads to a 60 % share of electricity generation by the RES (excluding large hydro).

Fig. 10 shows the hourly electricity and V2G demand in the 2030 HREP scenario. The net electricity demand across all sectors (except the transport sector) is 30.5 TWh/year, whiles the electricity demand of V2G is 1.62 TWh/year. The maximum electricity and V2G demand in the year 2030 for HREP are 4582 and 369 MW, respectively. Fig. 11 shows the hourly electricity production of the various power plants in the 2030 HREP scenario. The total electricity production reaches 45.37 TWh in 2030, of which solar and wind contribute 26.45 % and 30.86 % as opposed to their 2.36 % and 1.78 % contributions in the 2030 BAU, respectively. In the same vein, the contribution of thermal power reduces from 74 % in the 2030 BAU to 23.34 % in the 2030 HREP scenario. The hourly electricity consumption of the PHS pump and V2G and electricity production of the PHS turbine is shown in Fig. 12. The V2G and PHS pump consume 1.8 TWh/year and 0.26 TWh/year of electricity, respectively. On the other hand, the PHS turbine discharges 0.17 TWh/year of electricity. Finally, the hourly V2G and pumped hydro storages are shown in Fig. 13. The maximum and mean storage capacities of PHS and V2G are 33,000 and 2606 MW and 518 and 205 MW, respectively. In the current work, PHS and V2G are primarily included to provide system stability and flexibility by ensuring CEEP does not exceed 0. As such, the magnitude of charging and storage capacities of PHS and V2G will most likely follow the same trend as that of the average CEEP (>0) available every hour prior to the inclusion of these storage technologies. For instance, at 0.61 TWh CEEP (prior to the 2030 HREP, i.e., 2030 BAU with combined maximum penetrations of wind and solar), January, February, and August recorded the highest monthly average CEEP values (partly due to relatively low electricity demands in these months) and as such could explain the relatively higher charging and storage capacities of V2G and PHS in these months compared to the other months.

3.6. 2030 HREP costs

3.6.1. Will the proposed 2030 scenario (HREP) be financially feasible and competitive against the planned 2030 BAU?

Besides the 2030 HREP scenario's significant technical and environmental advantages over the 2030 BAU scenario, it is interesting to know that the entire system in 2030 HREP is relatively cheaper than the 2030 BAU. Since 2030 HREP was simulated based on technical simulation, the 2030 BAU was re-simulated (market economic simulation used in previous sections) based on technical simulation for a uniform basis of comparison. Table 10 presents the complete breakdown of the total costs involved in the two scenarios. The total annual cost for the 2030 HREP scenario is 11.48 % lower than the 2030 BAU case. It is seen that due to the inclusion of several new plans in the 2030 HREP scenario, the annual investment cost of the 2030 HREP scenario is 118 % higher than the 2030 BAU case. However, due to significant reductions in fuel consumption/natural gas exchanges and profit in electricity exchanges,

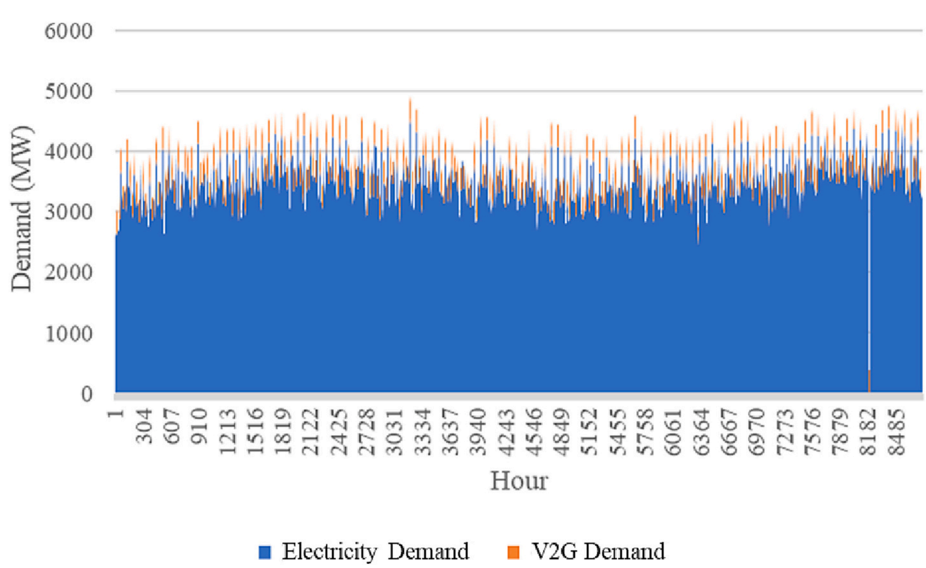


Fig. 10. Hourly values for electricity and V2G demand (MW) in 2030 HREP.

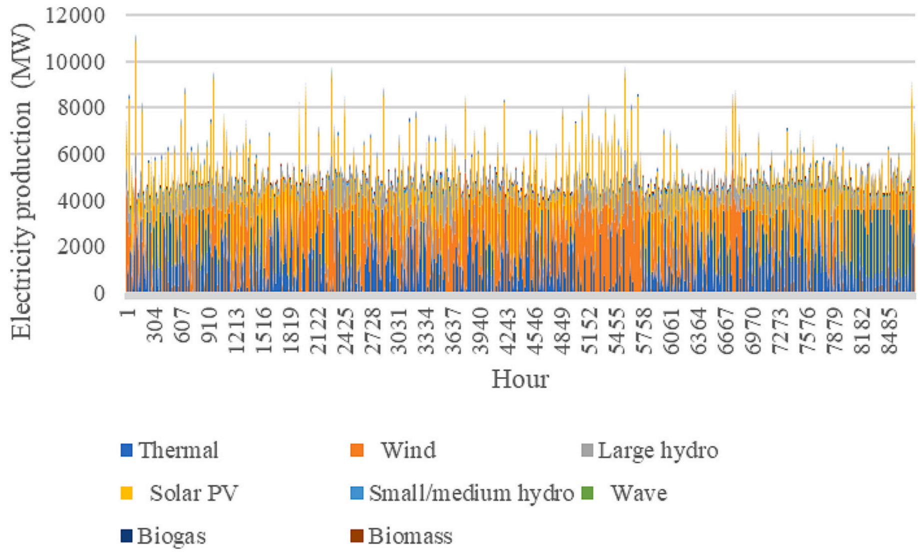


Fig. 11. Hourly values for electricity production of powerplants (MW) in the 2030 HREP scenario.

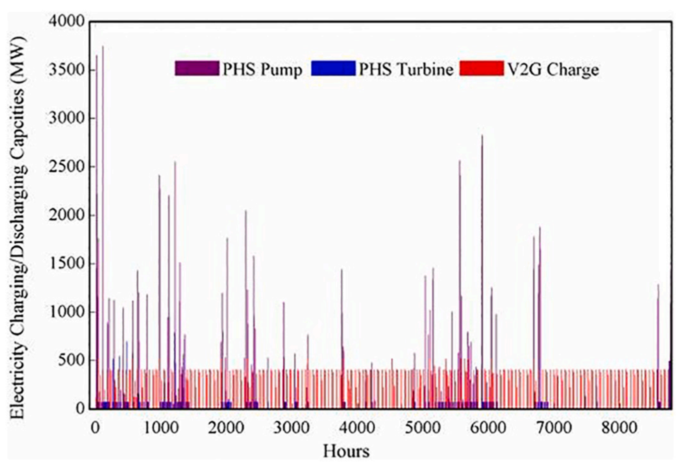


Fig. 12. Hourly values for electricity consumption of PHS pump and V2G, and electricity production of PHS turbine (MW) in the 2030 HREP scenario.

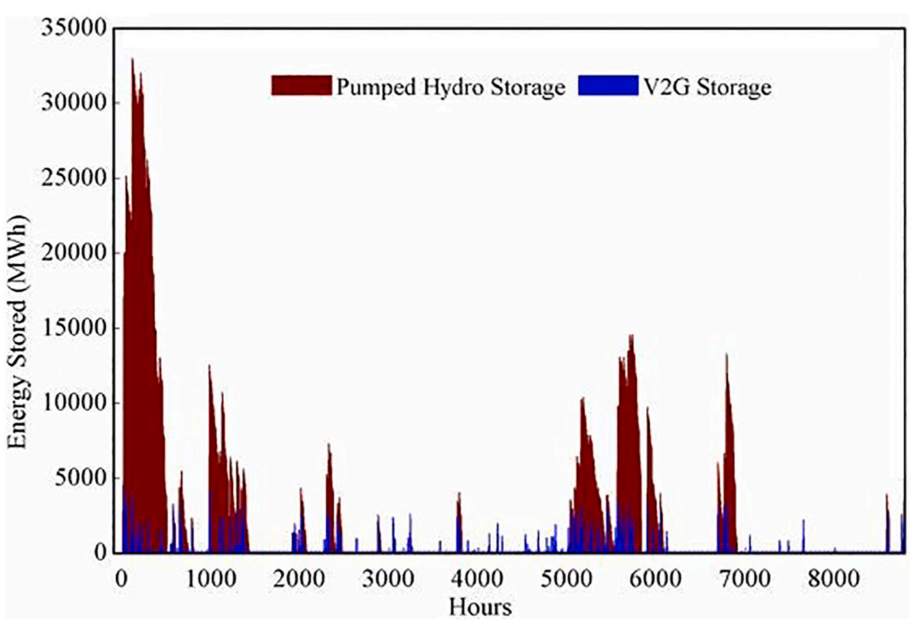


Fig. 13. Hourly values of electricity storage of PHS and V2G (MWh) in the HREP scenario.

Table 10
Economic comparisons between 2030 BAU and 2030 HREP.

	2030 BAU			2030 HREP		
	Total investment cost (MEUR)	Annual investment cost (MEUR)	Fixed O&M cost (MEUR)	Total investment cost (MEUR)	Annual investment cost (MEUR)	Fixed O&M cost (MEUR)
Components	25,728	1400	598	56,201	3051	1159
Variable costs (MEUR)						
Fuel oil		653			588	
Diesel		4687			4400	
Petrol/JP		2294			1880	
Gas handling		87			41	
Biomass		1272			1011	
Natural gas exchange		1955			828	
Marginal operation cost		70			29	
Electricity exchange		−1333			−2483	
CO_2 emissions cost		1425			1100	
Sub totals (MEUR)						
Variable costs		11,110			7394	
Fixed operation		598			1159	
Annual investments		1400			3051	
Total annual costs (MEUR)		13,108[a]			11,603	

[a] Cost here is based on technical simulation (for even comparison with 2030 HREP), whereas the total cost mentioned in Section 3.4 is based on economic simulation.

the HREP scenario produces an overall cheaper pathway to decarbonization than the BAU case. Due to the significant electricity exports, 2483 MEUR of electricity profits is possible in the HREP scenario compared to the 1333 MEUR in the BAU scenario. Natural gas exchange costs in the 2030 HREP will also be 57.65 % lower than the BAU case. Costs associated with fuel consumption/gas handling (excluding natural gas exchange) will also be 11.93 % lower in the HREP scenario compared to that of BAU. As such, the former's cost of CO_2 emissions is 22.81 % lower than the latter. The trend helps provide justification for the relatively lower total annual cost in the HREP scenario than in the BAU case despite adding several new fuels, technologies, and capacities.

In this section, we have shown that besides HREP scenario providing significantly higher technical penetration levels of wind and solar power and higher RE consumption than the planned 2030 BAU case, the proposed scenario in this study also provides better economic characteristics – thereby improving the overall feasibility of the 2030 HREP scenario.

Table 11 gives a quick snapshot of the comparison between the 2030 BAU case against the proposed 2030 HREP scenario based on the results obtained so far.

3.7. Discussion

Ghana's Renewable Energy Masterplan 2030 targets the creation of 220,000 jobs and 11 Mt avoided CO_2 emissions. Our analysis shows that the country stands a better chance of achieving and likely surpassing these targets in the proposed HREP 2030 scenario. For instance, with just solar and wind energy, a combined 26 TWh of electricity production at the maximum penetration of these two VRES could be realized in the HREP 2030 scenario. Given that solar PV and wind turbine technologies inherently have an employment potential of $0.278 * 10^{-7}$ /kWh/year [63], we estimate that the 26 TWh electricity production in the 2030 HREP scenario by these two VRES will generate 723 jobs compared to 47 jobs in the BAU case (utility-scale solar and wind). These values are significantly lower compared to 220,000 jobs, but the main takeaway is that when judged by the same metric or job potential estimations, our proposed pathway delivers a higher outcome than the BAU. The most recent data for CO_2 emission factor for solar and wind projects in Ghana is 0.46 tCO_2/MWh [1] - for the combined 26 TWh electricity produced by wind and solar energy in the HREP 2030 scenario, it will ultimately lead to an avoided CO_2 emissions of 11.96 Mt as opposed to the 0.78 Mt in the BAU case (utility-scale solar and wind).

To maximize RE consumption, it is proposed that the other energy-consuming sectors should not be overlooked. With the transport, industry, and residential sectors largely depending on carbon-based fuels, it is essential for the gradual introduction of biofuels, electric vehicles, hydrogen, and solar thermal technologies in the short term before 2030. This will usher a reference pathway for full decarbonization measures to be introduced in the longer term, at and beyond 2050. Given the same conditions as the 2030 BAU case, the total RES share of PES increases from 21.3 % to 37.2 % with the introduction of these zero or low-carbon measures by 2030 in the HREP scenario. However, if the export transmission line capacity should remain unexpanded than the predicted 2030 capacity, then only 4.23 TWh of electricity out of the 4.7 TWh

Table 11
2030 BAU versus HREP case.

Variable	2030 BAU	2030 HREP	Change of BAU case relative to HREP case (%)
CO_2 emissions (Mt)	41.18	31.80	−22.78
Renewable energy electricity production (TWh/year)	10.53	34.78	+230.29
Fuel consumption (TWh/year)	215.19	198.60	−7.71
Renewable energy share of electricity production excluding large hydro (%)	8	60	+650
Total annual cost (MEUR)	13,108	11,603	−11.48
Job creation by solar and wind (persons)	47	723	+1438.30
Avoided CO_2 emissions (Mt)	0.78	11.96	+1433.33

exportable electricity can be exported out of the country (prior to V2G and PHS inclusion) with 0.47 TWh of CEEP remaining in the system with the potential to cause a total shutdown to the entire operation of the power network. The government of Ghana is thus advised to consider at least one of the following two options if the current study's optimum technical penetration levels of solar and wind energy are to be maintained; (1) Expansion of export transmission line capacity. This option will ensure that the excess electricity that cannot be consumed domestically is exported to neighboring countries with lower electricity self-sufficiency rates. Though this could be a cheaper option, its success depends on the development of the neighboring countries and other political/diplomatic factors. For instance, if these close proximity neighbors increase their electricity self-sufficiency rate, that will imply a reduction in how much electricity can be exported, which could cause a potential non-zero CEEP remaining in the system. Furthermore, in the events of political tensions between Ghana and any of these countries, export could cease or reduce and thus expose the system to CEEP. (2) Introduction of storage technologies. Though relatively an expensive option, the inclusion of V2G, P2G and PHS would ensure that excess electricity is stored and, when required, discharged to serve the load (direct in the case of V2G and PHS but stationary fuel cell or dedicated gas turbines integration to re-electrify hydrogen in the case of P2G). V2G, P2G and PHS provide flexibility, storage, and matching of generation time to time of load, consequently eliminating CEEP.

One of the main drawbacks to high penetrations of solar and wind energy has to do with their land requirements. At 32.66 and 38.11 % technical penetrations of solar and wind energy, respectively, we estimate a total land requirement of 899 km^2 given that solar and wind energy power production requires 50 km^2 and 100 km^2 per 1 GW of installed capacity, respectively [64]. This will represent 0.4 % of the total land area of Ghana. In the longer term, as 100 % decarbonization becomes the order of the day, the land requirement will increase than the estimated 2030 levels since more solar and wind energy technologies would have to be installed. As such, we recommend the following three solutions in tackling the land requirement issues of high penetration rates of VRES; (1) floating solar PV used in conjunction with ground-mounted and roof-top solar PV, (2) offshore wind turbines used in conjunction with onshore wind turbines and (3) the capacity factor of the VRES technologies should improve significantly.

4. Conclusion

In the current study, an attempt has been made to investigate innovative ways of increasing variable renewable energy capacity without creating technical challenges to the electricity supply system. These innovative strategies include EV integration, power-to-hydrogen, and pumped hydro storage. The current study is set on Ghana as a case study area, with the aim of devising a better renewable energy plan for Ghanaian policymakers during the planning and development of long-term renewable energy targets beyond 2030. The main findings from the current study are summarized below.

(1) The current study has shown that as of the end of 2019, the technical penetration of solar energy was only 0.29 %. However, with the same existing conditions (i.e., demands and import/export transmission line capacity), the maximum technical penetration of solar energy that could have been explored in 2019, considering 0 CEEP, was 8.7 %.
(2) Based on the planned BAU target for 2030, our analysis shows that the technical penetration levels of solar and wind energy are only 2.64 and 1.99 %, respectively. Given the same conditions, non-zero CEEP can still be avoided at 32.66 and 38.11 % for solar and wind, respectively.
(3) We also estimate that the 26 TWh electricity production in the 2030 HREP scenario by wind and solar will generate 723 jobs compared to 47 jobs per year in the BAU case.
(4) For the combined 26 TWh electricity produced by wind and solar energy in the HREP 2030 scenario, it will ultimately lead to an avoided CO_2 emissions of 11.96 Mt as opposed to the 0.78 Mt in the BAU case (utility-scale solar and wind).

In summary, the proposed HREP pathway delivers the following benefits relative to the BAU case: (i) reduce CO_2 emissions by 23 %, increase renewable electricity production by 23 %, reduce fuel consumption by 8 %, reduce total cost for policy implementation by 11 %, and increases job potential and avoided CO_2 emissions by over 1400 %. Future studies can go beyond the investigation here to make a more detailed assessment of the land and water implications from the transitions we have proposed here using earth system models such as the Global Change Assessment Model.

CRediT authorship contribution statement

Jeffrey Dankwa Ampah: Writing - Original draft and Preparation, Software, Investigation, Formal analysis, Writing - Reviewing and Editing. **Sandylove Afrane**: Data curation, resources. **Bowen Li**: Validation. **Humphrey Adun**: Writing - Reviewing and Editing. **Ephraim Bonah Agyekum**: Writing - Reviewing and Editing. **Abdulfatah Abdu Yusuf**: Writing - Reviewing and Editing. **Olusola Bamisile**: Writing - Reviewing and Editing. **Haifeng Liu**: Supervision, Writing - Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This study was funded by Science Fund for Distinguished Young Scholars of Tianjin (No. 20JCJQJC00160) and by National Natural Science Foundation of China (No. 51921004).

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